

Image Compression using PSO-ALO Hybrid Metaheuristic Technique

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Abstract

Storing and transmission of images over the Internet have become a great concern nowadays keeping limited resources in hand. To conquer these issues, one of the solutions that have proved to be correct is image compression. Image compression techniques try to reduce the redundant information from an image with minimum loss of quality. In this paper, Ant lion optimization (ALO) and Particle swarm optimization (PSO) are hybridized as a proposed algorithm to compress an image of RGB subject. Both image encoding and decoding processes are simulated in our experiments. Results show that the compression ratio and PSNR of various images using the proposed algorithm are much higher than the ones using individual algorithms.

Keywords: ALO; hybrid metaheuristic technique; image compression; PSO

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1. Introduction

Compressing any subject leads to a reduction of consuming resources such as bandwidth and disk space. When compression is applied to an image, redundant or irrelevant data bits are dropped out and the image is reconstructed with an aim of minimum loss. Repetition of data in an image is generally classified as:

- **Coding Redundancy:** In this type of redundancy, color intensity values are replaced by bit codes that require less memory. Replacing with bit codes leads to compressing an image in terms of size.
- **Interpixel Redundancy:** This type of redundancy is also known as spatial & temporal redundancy. In an image, pixels are generally related to or similar to neighboring pixels, leading to replicas. Here, interpixel correlations are observed and treated.
- **Psychovisual Redundancy:** To visualize an image, all data in an image is not required. There exists some irrelevant data for visual perception and it can be discarded. Removing unnecessary/irrelevant information to see an image reduces its size for displaying as well as storage. This type of removing redundancy comes under lossy compression and lost data cannot be recovered. Quantization is the process used to reduce this type of redundancy in which large input values are mapped to a smaller number of output values.

In this paper, a hybrid metaheuristic algorithm is acquainted with streamlining issue. Two primary segments which legitimize the nature of the arrangement are investigation and abuse. Every calculation has its individual limit and it might focus on one or the other investigation or abuse. At the point when we crossover more than one calculation, it will assist with improving the nature of the arrangement by zeroing in on both investigation and misuse. Furthermore, one more significant property is the hole between investigation and misuse. If the gap is high, at that point it will lead to the helpless arrangement. In the event that the hole is less in nature, it will yield great nature of the arrangement. This significant property makes analysts more genuine in investigating hybrid metaheuristic algorithms. Image compression are classified as lossy and lossless [1]. Lossy compression techniques involve some sort of loss in the reformed image; in lossless compression techniques, the image is reconstructed without any loss. Detailed descriptions and differences among these two techniques are summarized in Table 1.

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Table 1. Difference between Lossy and Lossless Compression

Lossy Compression	Lossless Compression
This Technique includes some deficiency of information.	This Technique includes no deficiency of data.
Information that has been compacted utilizing this method can't be recuperated and recreated exactly.	If information has been (lossless) packed, the first information can be recuperated from the packed information.
Quality corrupted contrast with unique image source.	Quality continue as before with unique image.
High compression up to 50% of unique information record size.	It is 2:1 and the most is 3:1 proportion.
Sound and Image compression utilizes lossy compression.	Text compression utilizes lossless compression
More information can be obliged in channel.	Less information can be obliged in channel.
Example: (i) Telephone System, (ii) Video CD	Example: (i) Fax Machine (ii) Radiological Imaging

Image Compression Technique: Image compression is significant as it diminishes information repetition, which in turn reduces the memory consumed and the transmission transfer speed. Likewise, attributable to its lossless resources, a strategy for compression that coordinates DCT and Huffman coding method was introduced and has demonstrated its productivity. From the results received, the technique produced superior compression esteem, with an outwardly minor distinction between compacted images and unique images. Outstandingly, numerous techniques for compression are accessible today, but there is a need for a compression strategy that is quick, storage efficient, easy to implement, and client friendly.

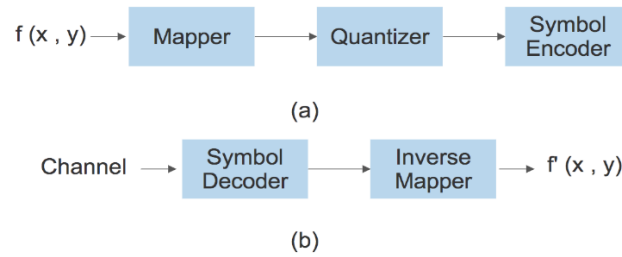


Figure 1. Image compression model: (a) Source encoder (b) Source decoder

As shown in Figure 1, image compression is concerned with the concept of removing the extra bits from an image keeping its quality maintained. The process of compression involves the representation of an original image into a reformed digital image with almost the same visual content but reduced size and quality. The need to preserve limited resources has given rise to the demand for decreasing computational costs. Compression helps in improving data storage as well as computational overhead. Several methods have been created for some real-world applications. These techniques focus on expanding effectiveness and profitability by thinking about these issues as an improvement. One methodology in tackling these improvement issues is by utilizing metaheuristic techniques. These techniques utilize some level of irregularity to look for an ideal arrangement, that is, they endeavor to rough the best answer for an issue utilizing an iterative experimental approach.

Population based metaheuristic techniques are mainly categorized as evolutionary algorithms, swarm-based intelligence algorithms, physics-based algorithms and bio-inspired algorithms. Evolutionary Algorithms (EA) are those algorithms that are based upon traditional sort of work culture. These algorithms are used to solve complex problems that are unsolvable in polynomial time. They follow the natural process of initialization, selection, genetic operators and termination. This process is repeated to create new generations with fitter members than the previous one until convergence is achieved. In recent years, many researchers are attracted to metaheuristics imitated by physical and biological processes of nature. Big Bang Crunch Optimization algorithm and gravitational search algorithm are good examples of physics metaheuristics. Biological based metaheuristic algorithms are based upon certain behavior of biological systems and use biology to solve complex problems in much less time and provide an approximate solution. Few bio-inspired metaheuristic techniques are Artificial neural network, evolutionary electronics, bio robotics and many more. Another most important and emerging field of population-based metaheuristics [2] is swarm intelligence (SI). Many SI algorithms have proven to solve optimization problems. It is based on two important concepts: self-organization and decentralization. Other SI algorithms include Genetic Algorithm, Ant Colony Optimization, Ant Lion Optimization, Particle Swarm Intelligence, Differential Evolution.

In this paper, the gap between exploration and exploitation is well understood, analyzed and implemented. This proposed hybridization is achieved by a slight change in formulas of ALO and parameters of PSO to improve compression parameters.

2. Related Methods and Scope

Ant Lion optimization (ALO) algorithm: Ant lion optimization algorithm, developed by Mirjalili in 2015, is based upon the unique hunting mechanism of ant lions for their prey. Ant Lions also known as doodlebugs, are famous for searching for solutions in diverse spaces. They have a life span of about 3 years which is divided into two phases: larvae and adult. They spent more than 90% of their life as larvae hunting; the rest of the is dedicated to reproduction. These antlion larvae hunt by digging a cone shape pit in the sand, hiding at the end of the cone. As the prey is trapped in the cone pit, they try to catch it in their jaw. Once caught, they consume it and prepare the pit for the trap of new prey. ALO algorithm works in the same manner as antlions hunt for their prey. Ants move in their search space and antlions wait for them. A random walk is chosen for ants initially to explore and its position is used as solution parameters. A fitness function is used to calculate fitness of all ants as well as antlions, as we assume antlions are also present in search space. Certain conditions are mentioned in [3] to trap an ant in an antlion pit. A roulette wheel is used to implement antlions hunting capacity by selecting more fit antlions during optimization to catch ants. This helps to eliminate falling in local optima too. To fall into the trap, mathematically ant's search radius is decreased adaptively, and updated position is saved. This guarantees the exploitation of search space. After the ant has fallen into the pit, antlion's position must be changed to the ant's latest position to catch it. With theoretical background just given, ALO algorithm can be defined as follows:

ALO algorithm
• Initialize parameters of ALO • Set the initial position of ants and antlion • Calculate fitness of each ant and antlion using objective function • While termination criteria is not met ○ Find elite antlion ○ For each ant ▪ Select antlion using roulette wheel ▪ Update antlion position using random walk ▪ Calculate fitness value of ants ▪ If ant is fitter • Update antlion's position to ant's latest position ▪ If antlion is fitter than elite • Update elite to antlion • Return elite's position as optimized solution

Particle Swarm Optimization (PSO): Particle Swarm Optimization (PSO), created in 1995 by Dr. James Kennedy, is one of the most commonly used heuristic algorithms. It is population-based metaheuristic technique that mimics the behavior of flocking of birds or schooling fishes. PSO alludes to a moderately new group of calculations that might be utilized to discover ideal (or near optimal) answers for mathematical and subjective issues. It is easily executed in most programming dialects and has demonstrated both exceptionally viable and fast when applied to an assorted arrangement of advancement issues. In this paper, PSO implemented is inspired by a group of birds looking for their food. Each bird or particle has an idea about the distance to the food or to the solution. All birds/particles are evaluated by the fitness function to be optimized. To find the global minimum solution, each bird/particle moves with a certain velocity and updates its position over many iterations to achieve near solution. Over the iterations, each particle updates its positions and direction based upon its previous best position (local best). Along with its local best position, it also takes care of the best solution of the group (global best).

PSO Algorithm
• Randomly initialize group of population particles • Initialize velocity & position • While termination criteria is not met ○ For each particle ▪ Compute position as pBest ▪ Compute velocity ▪ If current position of particle is better than pBest • Assign current position to pBest ▪ Else • Assign pBest to gBest ○ Update particle position for next generation

3. Literature Review

Kai-jen Cheng et al. [1] have proposed Binary Embedded Zero Tree Wavelet calculation which is identified with the recently characterized two tree structures. In BEZW calculation, the spot streams are orchestrated all together depending on their significance, with the goal that the reproduction deviation be dependent upon the arrangement of recuperated bit planes. In 3-D-BEZW calculation, the overall pass accomplishes the quantization and noteworthiness test.

Priya Pareek et al. [4] have suggested RLE as an appropriate technique for compacting any type of information independent of the data mollified. Be that as it may, elective RLE plans can encode information along the length of the bitmap, and the sections are to encode the bitmap as 2-D tiles and to encode pixels in the inclining structure are made into a crisscross structure.

Authors, Sophin Seeli et al. [5] and K uppusamy et al. [6] have recommended that compressing an image by diving into small fractals is a very fruitful procedure. It is because some parts of an image are similar to each other and can be replaced by a single block. This helps in minimizing the memory requirement for an image. Image blocks are stored as mathematical equations rather than color intensities. Iterated function systems codes are used to replace image blocks with similar ones. This type of compression works well when the image is regular and has self-similarity properties.

Tahamohammed Hasan et al. [7] have introduced an Adaptive Fractal Image Compression (AFIC) technique to decrease the high encoding time in the Fractal Image Compression (FIC). AFIC is utilized to improve encoding time and achieve better compression ratio with minimum quality loss in reformed images. In correlation with certain techniques, AFIC invests considerably less encoding energy and gives a high compression ratio of the nature of the remade images.

K. S. Lee and Z. W. Geem et al. [8] have developed modified Harmony Search (HS) for more complex real world optimization problems. Authors believe that when there are more than one local optima, the final solution always depends on initial values taken which may or may not give the global optimum solution. Modified HS deals with continuous design variables and is based on the musical process to find a perfect harmony state.

F. Saitoh et al. [9] considered image improvement as a programmed defined cycle. A defined change work is utilized to upgrade the image, which is then assessed by a target wellness rule. The part of the advancement procedure utilized is to expand the target assessment work by finding the ideal design of boundaries of the change work. Programmed image upgrade is a perplexing improvement issue with an enormous issue example, requiring the utilization of metaheuristic methods. In a previous study, few metaheuristic calculations have been utilized for this reason.

C. Blum, M. J. Blesa Aguilera, et al. [10] In this paper, we present an examination of the presentation of cross breed metaheuristic and old style metaheuristic in the field of an image upgrade. Cross breed metaheuristic is a term instituted to speak to a class of calculations that utilize a blend of metaheuristic with some other improvement method such as Dynamic Programming, Integer Linear Programming (ILP) or other metaheuristic calculations.

Metin Toz in [11] proposed an improved ALO (IALO) to form clusters of an image. The author has used a boundary decreasing procedure to enhance ALO objective function. The objective function used to cluster an image is modified to provide better results. The proposed algorithm was implemented on 26 raw mathematical equations and different objective functions. Results show that IALO yields better results than ALO, chaos-ALO, PSO and many more metaheuristic algorithms.

Malik Braik, AlaaSheta and Aladdin Ayeshet al. in [12] have used PSO to enhance an image. This is done using new automatic enhancement techniques including swarm algorithms. The improvement cycle is a non-straight streamlining issue with a few requirements. The goal of the proposed PSO is to amplify a target wellness rule to improve the contrast and detail in an image by adjusting the boundaries of a novel expansion to a neighborhood upgrade strategy. The achievability of the proposed strategy is exhibited, contrasted and compared with Genetic Algorithms (GAs) based image improvement procedure. Results show that PSO outperforms GA in computational time and enhanced resultant image.

4. Proposed Method

The main aim of the method proposed is to compress the image at a considerably high compression ratio without degrading the quality. It helps in the reduction of memory requirement and transmission cost. In past, metaheuristic techniques have proven to give a better approximate solution to most of the combinatorial problems. Here, two metaheuristic techniques are hybridized together to achieve the compression goal. The proposed method combines PSO and ALO using low-level evolutionary mixing method. Both techniques are used in parallel for the final algorithm to produce fitter members for next

generation based on distinct values. Exploitation of the members is enhanced using PSO and ALO is used to increase the exploring rate in the search space. Detailed proposed hybrid method is well explained in the flowchart in Figure 2.

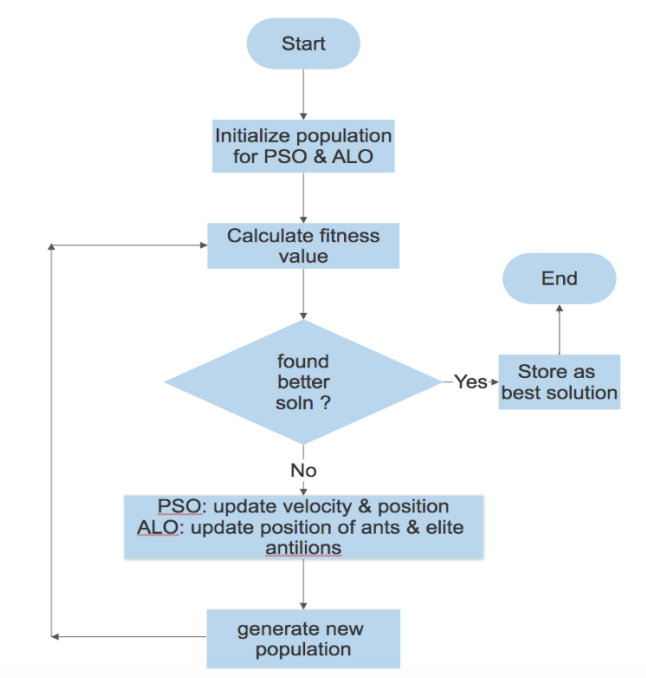


Figure 2. Proposed Architecture for Hybrid PSO-ALO Algorithm

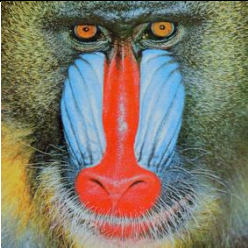
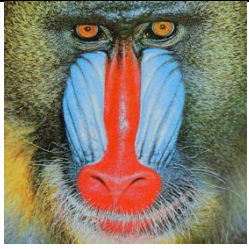
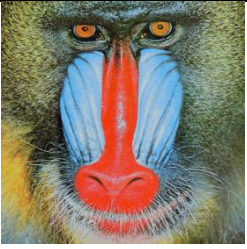
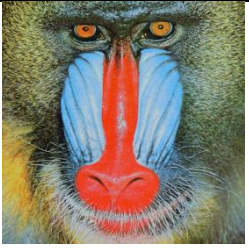
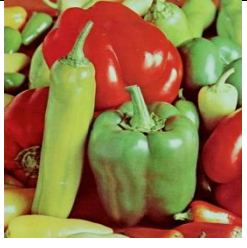
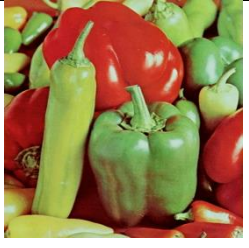
Original Image	Compressed Image using PSO	Compressed Image using ALO	Compressed Image using proposed hybrid technique (ALO-PSO)
 lena.jpg			
 mandril.png			
 peppers.png			

Figure 3. Experimental Results showing original image and compressed images

5. Results and Discussion

Our proposed technique is implemented using MATLAB (version 18a) On the basis of hybrid ALO and PSO algorithm the intended image compression technique is evaluated by using images. The performance of our proposed technique is accessed by means of performance metrics. Effective image compression is done by using a hybrid algorithm as shown in Figure 3.

The results have been compared to the existing metaheuristic algorithm mentioned in the table in terms of encoding time and PSNR of image compression using a hybrid metaheuristic algorithm.

Table 2. Performance Analysis

Image	Method	Original Size (kb)	Compressed Image (kb)	Compression Ratio	PSNR (dB)	Encoding Time(s)
Lena.jpg	Proposed (PSO-ALO hybrid)	592	27	22	41.3	27.48
	PSO	592	47	13	34.45	36.79
	ALO	592	43	14	29.95	49.58
mandril.png	Proposed (PSO-ALO hybrid)	633	205	4	42.45	27.56
	PSO	633	208	4	41.32	38
	ALO	633	220	3	42.3	55
peppers.png	Proposed (PSO-ALO hybrid)	544	142	4	50.46	27.45
	PSO	544	142	4	40.1	35.12
	ALO	544	153	4	46.43	42

Table 2 illustrates that the performance of our proposed algorithm and is compared with traditional optimization techniques: PSO and ALO. Performance is compared based on parameters like PSNR and Encoding Time. It reveals that our proposed algorithm has shown higher values for the parameters like PSNR and Encoding Time. Therefore, it clearly shows that our proposed technique yields higher image compression results than the traditional techniques. Figure 4 illustrates the comparison graph of our proposed algorithm with traditional optimization techniques.

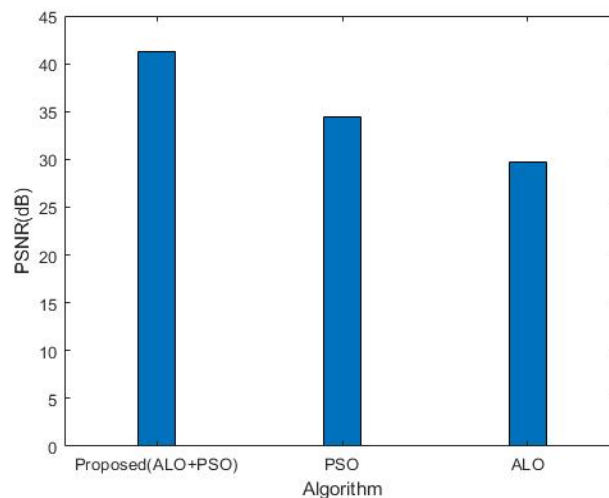


Figure 4. Performance Graph among 3 techniques

6. Conclusion and Future Scope

Limited availability of resources for storage and transmission purpose has given rise to compressing an image. Various traditional lossy and lossless compression techniques have been explored and it has been noted that to solve the NP problem, metaheuristic approach gives much better results. In this paper, two metaheuristic techniques are hybrid together to increase the compression ratio. The goal of the proposed algorithm is to highlight the pixels in the edges to visualize the image more properly. When in doubt, the consequence of hybridization can make a couple of upgrades similar to either computational speed or precision. Three sample images are compressed using the proposed hybrid algorithm and individual ones. Capable measures such as compression ratio, PSNR and encoding time, are used to do the assessment of the reconstructed image.

According to the results obtained, the proposed algorithm gives high compression ratio and PSNR values as compared to individual techniques.

Based on the analysis done and results obtained, there is a chance of improvement in compression metrics. Further, examinations might be performed by altering hybrid parameters to speed up the compression process. In near future, the proposed algorithm can also be used for high resolution image analysis such as band selection, compression or image enhancement.

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